DATA REDUCTION

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What is a Data Reduction?

- Data reduction can be applied to obtain a reduced representation of the dataset that is much smaller in volume, yet maintains the integrity of the original data.
- It is the condensed description of the original data which is much smaller in quantity but keeps the quality of the original data.
- Not only do the data need to be condensed for the sake of manageability, they also have to be transformed so they can be made intelligible in terms of the issues being addressed.
- It means large amounts of data are cleaned, organized and categorized based on prerequisite criteria to help in driving business decisions.
- The purpose of data reduction can be two-fold:
  - Reduce the number of data records by eliminating invalid data; or
  - Produce summary data and statistics at different aggregation levels for various applications.

Methods of Data Reduction

- Data cube aggregation
  - Data Cube Aggregation is a multidimensional aggregation that uses aggregation at various levels of a data cube to represent the original data set, thus achieving data reduction. Data Cube Aggregation, where the data cube is a much more efficient way of storing data, thus achieving data reduction, besides faster aggregation operations.
  - E.g. Year (2020) sales Data aggregation Method
    - Q 1 20 Crore Sale in year 2020 90 Crore
    - Q 2 30 Crore
    - Q 3 40 Crore

- **Dimensionality Reduction**
- Dimensionality Reduction is the process of reducing the number of dimensions the data is spread across. It means, the attributes or features, that the data set carries as the number of dimensions increases the sparsity. This sparsity is critical to clustering, outlier analysis and other algorithms. With reduced dimensionality, it is easy to visualize and manipulate data
- Lossy data: it reduces a file by permanently eliminating certain information.
- Lossless data: Original data can be recovered when the file in uncompressed.
- **Numerosity Reduction**
- This method uses alternate, small forms of data representation, thus reducing data volume. There are two types of Numerosity reduction, Parametric and Non-Parametric.

- Parametric data reduction:
  - A parametric data reduction technique is a data reduction technique that assumes a certain model for the data. The model contains some parameters and the technique fits the data into the model to determine the parameters. Then data reduction can be performed.
- Non-parametric data reduction
  - A non-parametric numerosity reduction technique does not assume any model. The non-Parametric technique results in a more uniform reduction, irrespective of data size, but it may not achieve a high volume of data reduction like the Parametric one. There are at least four types of Non-Parametric data reduction techniques, Histogram, Clustering, Sampling, Data Cube Aggregation, Data Compression.

- Data compression:
  - Data compression is the process of encoding, restructuring or otherwise modifying data in order to reduce its size. Fundamentally, it involves re-encoding information using fewer bits than the original representation.

Process of data reduction

- **Data Cleaning**:
  - **Identify and Handle Missing Values**: Use Transform > Replace Missing Values or manually handle missing values by coding them as specific values or using mean/mode imputation.
  - **Identify and Correct Errors**: Use descriptive statistics and visualizations to spot outliers or errors.
- **Data Transformation**:
  - **Recode Variables**: Use Transform > Recode into Same Variables or Recode into Different Variables to categorize or transform data.
  - **Compute Variables**: Use Transform > Compute Variable to create new variables from existing ones.

- **Data Reduction Techniques**:
- **Factor Analysis**:
  - Use Analyze > Dimension Reduction > Factor to reduce the number of variables by identifying underlying factors.
  - Choose extraction methods like Principal Component Analysis (PCA) and rotation methods (Varimax, Promax).
- **Cluster Analysis**:
  - Use Analyze > Classify > K-Means Cluster or Hierarchical Cluster to group similar cases or variables into clusters.

- **Feature Selection**:
  - **Correlation Analysis**: Use Analyze > Correlate > Bivariate to identify and eliminate highly correlated variables.
  - **Stepwise Regression**: Use Analyze > Regression > Linear and select stepwise methods to include only significant predictors.
- **Data Summarization**:
  - **Descriptive Statistics**: Use Analyze > Descriptive Statistics > Frequencies, Descriptives, or Explore to summarize data.
  - **Crosstabs**: Use Analyze > Descriptive Statistics > Crosstabs to examine relationships between categorical variables.
- **Data Visualization**:
  - **Charts and Graphs**: Use Graphs menu to create visual representations like histograms, boxplots, and scatterplots to identify patterns and reduce complexity.